

MedTrack_DV

Dashboard Testing & Validation Report

Milestone 4 – Module 7: Testing & Validation

Project: Hospital Operations & Patient Analytics Dashboard

Executive Summary

This report documents the testing and validation performed for the MedTrack_DV Hospital Operations & Patient Analytics Dashboard. The testing covered the four analytical dashboards—Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization—as well as the About and Help pages. KPI calculations, healthcare metrics, patient-flow analytics, filters, navigation, and dashboard interactions were reviewed. Based on the testing performed, all reported areas were functioning correctly and no major dashboard issues were identified.

1. Project Overview

MedTrack_DV is designed to transform hospital operational and patient data into actionable insights for monitoring patient flow, department performance, hospital performance, and healthcare resource utilization. The project specification defines a unified dashboard suite covering Hospital Overview, Patient Flow, Department Analytics, and Resource Utilization.

2. Testing Objectives

The objectives of Module 7 testing were to validate KPI calculations, verify healthcare metrics, test dashboard interactions, and validate patient-flow analytics. The project specification sets the evaluation expectation of no major dashboard issues and KPI accuracy above 95%.

3. Testing Scope

Test Category	Components Covered	Result
KPI Validation	Hospital KPIs and calculated metrics	PASS
Hospital Overview	Admissions, occupancy, readmission, trends	PASS
Patient Flow	Admissions, discharges, movement, average stay, peak load	PASS
Department Analytics	Performance, volume, readmission, efficiency, capacity	PASS
Resource Utilization	Beds, staff, equipment, capacity, availability	PASS
About Page	Page access, content and navigation	PASS
Help Page	Page access, guidance and navigation	PASS
Interaction Testing	Filters, actions and dashboard interactions	PASS
Navigation Testing	Page-to-page navigation and dashboard linking	PASS

4. Testing Methodology

Testing was performed through functional checks of the completed dashboard. Each component was opened and reviewed to confirm that it displayed the intended information. KPI results were checked for correctness; filters and interactive controls were operated to verify expected responses; navigation controls were tested between pages; and patient-flow visuals were reviewed for correct analytical behavior. The About and Help pages were also checked as part of the complete dashboard experience.

5. KPI Validation

KPI / Metric	Validation Focus	Result
Total Admissions	Value and display consistency	PASS
Occupancy Rate	Calculation and displayed result	PASS
Average Length of Stay	Calculation and displayed result	PASS
Readmission Rate	Calculation and displayed result	PASS
Bed Utilization Rate	Calculation and displayed result	PASS
Department Efficiency Score	Calculation and department comparison	PASS

The project specification identifies these six KPIs for hospital KPI engineering. All listed KPI areas were reported as working correctly during validation.

6. Dashboard Functional Testing

Dashboard	Testing Performed	Expected Behavior	Result
Hospital Overview	KPI cards, admissions, occupancy, readmission and trends reviewed	Relevant hospital performance information displays correctly	PASS
Patient Flow	Admission, discharge, movement, average stay and peak-load visuals reviewed	Patient-flow analytics display and respond correctly	PASS
Department Analytics	Department performance, volume, readmission, efficiency and capacity reviewed	Department comparisons and metrics display correctly	PASS
Resource Utilization	Bed, staff, equipment, capacity and availability reviewed	Resource metrics and utilization insights display correctly	PASS

7. About and Help Page Testing

Component	Test Performed	Result
About	Opened the About section and checked that project information/content displays correctly.	PASS
Help	Opened the Help section and checked that guidance/information displays correctly.	PASS
Navigation	Tested movement to and from the supporting pages using the available navigation controls.	PASS

8. Filter, Interaction and Navigation Testing

Test ID	Function	Validation	Result
INT-01	Filters	Filter selections update relevant dashboard visuals correctly.	PASS
INT-02	Navigation	Navigation controls move between dashboard pages correctly.	PASS
INT-03	Dashboard Actions	Interactive dashboard controls respond as expected.	PASS

Test ID	Function	Validation	Result
INT-04	Dashboard Linking	Linked pages/components open correctly.	PASS
INT-05	Visual Response	Visuals remain consistent after interaction.	PASS

9. Patient Flow Validation

Patient-flow validation covered the project requirements of admission trends, discharge tracking, patient movement analysis, average stay analysis, and peak patient-load monitoring. These areas were reviewed and reported as functioning correctly.

Patient Flow Area	Validation Result
Admission Trends	PASS
Discharge Tracking	PASS
Patient Movement Analysis	PASS
Average Stay Analysis	PASS
Peak Patient Load Monitoring	PASS

10. Defect and Issue Summary

Issue Category	Issues Found	Status
KPI / calculations	0	No issue identified
Visualizations	0	No issue identified
Filters / interactions	0	No issue identified
Navigation / linking	0	No issue identified
About / Help pages	0	No issue identified
Patient-flow analytics	0	No issue identified
Major dashboard issues	0	No major issue identified

11. Overall Test Result

Validation Measure	Result
KPI validation	PASS
Healthcare metric validation	PASS
Dashboard functionality	PASS
Patient-flow validation	PASS
Interaction and navigation testing	PASS
About and Help testing	PASS
Major issues	None identified
Overall status	PASSED

12. Conclusion

The MedTrack_DV dashboard suite successfully completed the reported Module 7 testing and validation activities. The four analytical dashboards, supporting About and Help pages, KPI calculations, patient-flow analytics, filters, interactions, and navigation were tested and found to be functioning correctly. No major dashboard issues were identified. The project is therefore ready to proceed to Module 8: Documentation and Project Delivery.

Testing Status: PASSED